

16_Hessian Matrix, Convexity & Optimization

Hessian Matrix

The **Hessian matrix** of a scalar-valued function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is a square matrix of second-order partial derivatives. It describes the **local curvature** of the function. Formally:

$$H(f) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix} \quad H(f) = \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} & \frac{\partial^2 f}{\partial x \partial z} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} & \frac{\partial^2 f}{\partial y \partial z} \\ \frac{\partial^2 f}{\partial z \partial x} & \frac{\partial^2 f}{\partial z \partial y} & \frac{\partial^2 f}{\partial z^2} \end{bmatrix}$$

- $H(f)$ is **symmetric** if the second derivatives are continuous ($\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i}$).
- Each entry measures how the slope in one direction changes as you move in another direction.

Use in Optimization

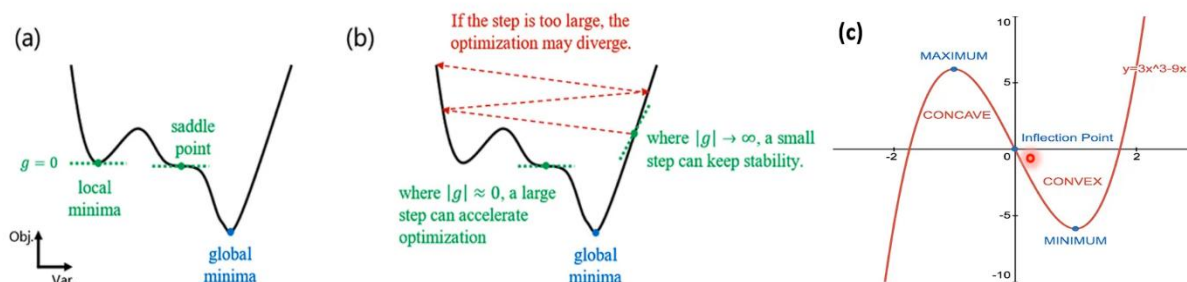
The Hessian is used to analyze **critical points** of a function:

1. **Local Minimum:** If $H(f)$ is **positive definite** at a point, f has a local minimum there.
2. **Local Maximum:** If $H(f)$ is **negative definite**, f has a local maximum.
3. **Saddle Point:** If $H(f)$ is **indefinite** (some eigenvalues positive, some negative), the point is a saddle point.

Role in Machine Learning

- Used in **Newton's method** and other second-order optimization techniques.
- Helps determine **convergence speed** and direction for updating parameters.
- If a function is **strictly convex**, then any local minimum is also a **global minimum**, and the Hessian is positive definite everywhere.

Convex and non-convex



- If a function $f(x)$ is **concave**, then $-f(x)$ is **convex**.
- Similarly, if $f(x)$ is **convex**, then $-f(x)$ is **concave**.

The function is **strictly convex**, and **any local minimum is also a global minimum**

1. Hessian matrix

The **Hessian** is a square, symmetric matrix of **second-order partial derivatives** of a scalar-valued function.

For a function $f(x_1, x_2, \dots, x_n)$:

$$H(f) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}$$

- **Positive definite Hessian** → local minimum.
- **Negative definite Hessian** → local maximum.
- **Indefinite Hessian** → saddle point.

For a **single-variable function** $f(x)$, the Hessian is just the second derivative:

$$H(f) = \frac{d^2 f}{dx^2}$$

2. Example: $f(x) = x^2$

Step 1: First derivative

$$f'(x) = \frac{d}{dx}(x^2) = 2x$$

Step 2: Critical point

Set $f'(x) = 0$:

$$2x = 0 \Rightarrow x = 0$$

Step 3: Second derivative (Hessian)

$$f''(x) = \frac{d^2}{dx^2}(x^2) = 2$$

So the Hessian is:

$$H(f) = [2](1 \times 1 \text{ matrix})$$

3. Determine nature of critical point

- $H(f) = 2 > 0 \Rightarrow$ **positive definite**
- Therefore, $x = 0$ is a **local minimum** (and also global, since x^2 is convex).

Example :

Function of $f(x, y) = x^2 + y^2$

$$H = \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \quad \text{Final Matrix}$$

Let's see step by step:

1st Order derivative:

$$\frac{\partial f}{\partial x} = \frac{d}{dx}(x^2 + y^2) = 2x$$

$$\frac{\partial f}{\partial y} = \frac{d}{dy}(x^2 + y^2) = 2y$$

2nd Order: $\frac{\partial^2 f}{\partial x^2} = \frac{d}{dx}(2x) = 2$ $\frac{\partial^2 f}{\partial y^2} = \frac{d}{dy}(2y) = 2$

$$H(f) = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

Let's Calculate the Eigenvalue $\det(H - \lambda I) = 0$

$$\begin{vmatrix} 2 - \lambda & 0 \\ 0 & 2 - \lambda \end{vmatrix} = 0$$

$$(2 - \lambda)(2 - \lambda) = 0$$

$$\lambda_1 = 2, \quad \lambda_2 = 2$$

- Since both eigenvalues are positive ($\lambda_1, \lambda_2 > 0$), the Hessian matrix is positive definite.
- A positive definite Hessian confirms that $f(x, y)$ is strictly convex.

Hessian Test For Local Min/Max & Convexity

Hessian Type	Eigenvalues λ	Local Min/Max Classification	Convexity Classification
Positive Definite	$\lambda_1 > 0, \lambda_2 > 0$	Local Minimum	Strictly Convex
Negative Definite	$\lambda_1 < 0, \lambda_2 < 0$	Local Maximum 	Strictly Concave
Indefinite	$\lambda_1 > 0, \lambda_2 < 0$	Saddle Point (Neither Min nor Max)	Non-Convex
Positive Semi-Definite	$\lambda_1 \geq 0, \lambda_2 \geq 0$	Possibly Local Minimum (Need Higher-Order Test)	Convex (Possibly Flat in Some Directions)
Negative Semi-Definite	$\lambda_1 \leq 0, \lambda_2 \leq 0$	Possibly Local Maximum (Need Higher-Order Test)	Concave (Possibly Flat in Some Directions)

The function is **strictly convex**, and **any local minimum is also a global minimum**